Test 1 curve:

Average: 29/60

A: 32 +

B: 22 - 32

C: 21 —

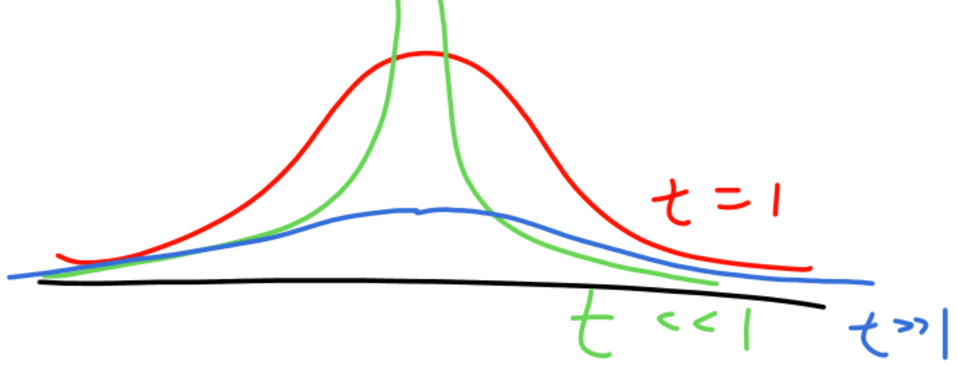
Heat Equation

$$\begin{cases} u_t - \Delta u = 0 & \text{in } (0,\infty)\times\mathbb{R}^n \\ u(x,0) = g(x) \end{cases}$$

Last time: Fundamental Solution / Heat kernel

$$\Phi(t,x) = \frac{1}{(4\pi t)^{n/2}} e^{-|x|^2/4t}$$

$$\rightarrow \int_{\mathbb{R}^n} \Phi(t,x)\,dx = 1 \quad \text{for any } t$$



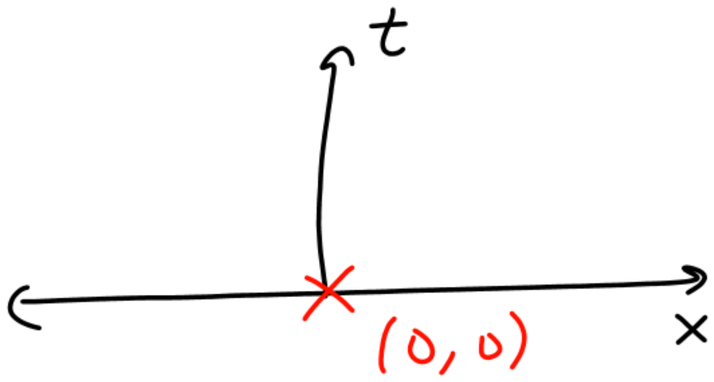
Claim:

$$\boxed{u = \Phi(t,x) *_x g(x)}$$

solves $u_t - \Delta u = 0$,

and $\lim_{t\to 0+} u(t,x) = g(x)$

as long as $g \in C(\mathbb{R}^n)$.



$\Phi(t,x)$ defined pointwise when $(t,x) \neq (0,0)$.

$$u(x,t) = \int_{\mathbb{R}^n} \Phi(t, x-y)\, g(y)\, dy$$

Last time: u solves

$$u_t - \Delta u = 0.$$

Assume g is bounded.

Proof that $u(x,t) \to g(x)$ as $t \to 0^+$:

Let $x_0 \in \mathbb{R}^n$, $\varepsilon > 0$ be given.

Choose $\delta > 0$ s.t.

$|g(x) - g(x_0)| < \varepsilon$ if $|x - x_0| < \delta$.

$$u(x,t) - g(x_0)$$

$$= \int_{\mathbb{R}^n} \Phi(t, x-y)\, g(y)\, dy - g(x_0)$$

$$= \int_{\mathbb{R}^n} \Phi(t, x-y)\, g(y)\, dy - g(x_0) \int_{\mathbb{R}^n} \Phi(t, x-y)\, dy$$

$$|u(x,t) - g(x_0)|$$

$$= \left| \int_{\mathbb{R}^n} \Phi(t, x-y)\left[g(y) - g(x_0)\right] dy \right|$$

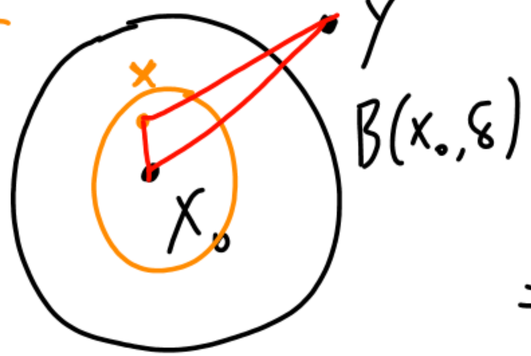
$$\Phi(t, x-y) = \frac{1}{(4\pi t)^{n/2}} e^{-|x-y|^2/4t}$$

$$\leq \int_{\mathbb{R}^n} \Phi(t, x-y) \, |g(y) - g(x_0)| \, dy$$

$$= \int_{B(x_0,\delta)} \Phi(t, x-y) \, |g(y) - g(x_0)| \, dy + \int_{\mathbb{R}^n \setminus B(x_0,\delta)} \Phi(t, x-y) \, |g(y) - g(x_0)| \, dy$$

$$= I_1 + I_2$$

$|x - x_0| < \frac{\delta}{2}$



$$\delta \leq |x_0 - y| \leq |x_0 - x| + |x - y|$$

$$\leq \frac{\delta}{2} + |x - y|$$

$$\Rightarrow |x - y| \geq \delta/2$$

$$I_1 = \int_{B(x_0,\delta)} \Phi(t,x-y)\,|g(y)-g(x_0)|\,dy$$

$$< \int_{B(x_0,\delta)} \Phi(t,x-y)\,\varepsilon\,dy$$

$$\leq \varepsilon \int_{\mathbb{R}^n} \Phi(t,x-y)\,dy = \varepsilon$$

For $|x-x_0| < \delta/2$,

by above, $|x-y| \geq \delta/2$ when

$y \in \mathbb{R}^n \setminus B(x_0,\delta)$

$y = x - \sqrt{4t}\,z$

$$\Rightarrow \Phi(t,x-y) = \frac{1}{(4\pi t)^{n/2}} e^{-|x-y|^2/4t}$$

$$\leq \frac{1}{(4\pi t)^{n/2}} e^{-(\delta/2)^2/4t}$$

$$= \frac{1}{(4\pi t)^{n/2}} e^{-\delta^2/8t}$$

$$I_2 = \int_{\mathbb{R}^n \setminus B(x_0,\delta)} \frac{1}{(4\pi t)^{n/2}} e^{-|x-y|^2/4t}\,|g(x_0)-g(y)|\,dy$$

$$z = \frac{x-y}{\sqrt{4t}} \qquad dz = \frac{dy}{(4t)^{n/2}}$$

$$I_2 = \frac{1}{\pi} \int_{\mathbb{R}^n \setminus B\left(\frac{x-x_0}{\sqrt{4t}}, \frac{\delta}{\sqrt{4t}}\right)} e^{-|z|^2}\,|g(x_0) - g(x_0+\sqrt{4t}\,z)|\,dz$$

$$|I_2| \leq \int_{\mathbb{R}^n \setminus B\left(\frac{x-x_0}{\sqrt{4t}}, \frac{\delta}{\sqrt{4t}}\right)} e^{-|z|^2} |g(x_0) - g(x_0 + \sqrt{4t}\, z)| \, dz \longrightarrow 0$$

by Dominated Convergence Theorem

$$\lim_{t \to 0+} |u(x,t) - g(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \frac{\delta}{2}$$

Dominated Convergence Theorem:

If $f_n \longrightarrow f$ pointwise,

and $|f_n| \leq h$ with $\int_X |h| \, dx < \infty$,

then $\int_X f_n \, dx \longrightarrow \int_X f \, dx$.

Infinite Propagation Speed:

Assume $g \geq 0$, and

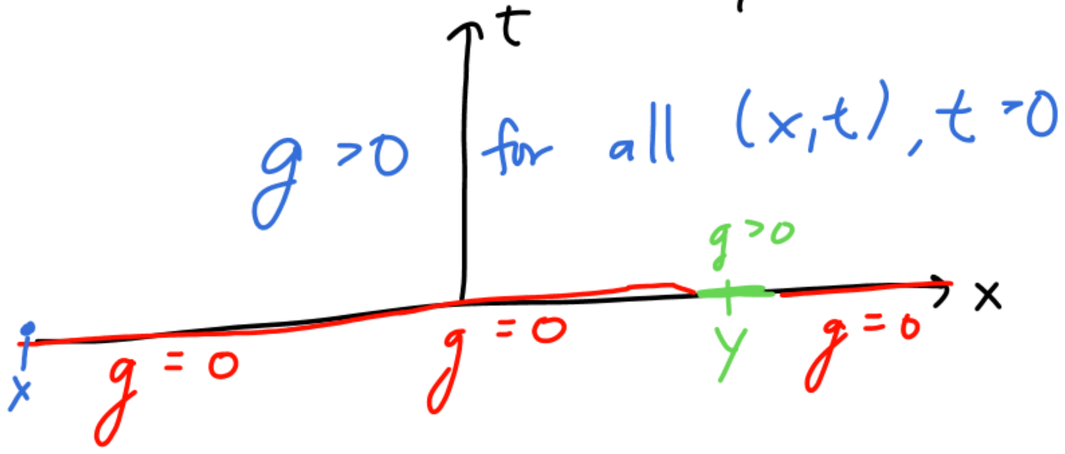
$u_t - \Delta u = 0$, $u(x,0) = g(x)$

Fundamental Solution gives a solution of the form

$$u(x,t) = \int_{\mathbb{R}^n} \Phi(t, x-y) g(y)\, dy$$

If $g = 0$ except in $B(x_0, \delta)$ but $g(x) > 0$ in $B(x_0, \delta)$,

we still have $u(x,t) > 0$ everywhere



Nonhomogeneous Heat Equation:

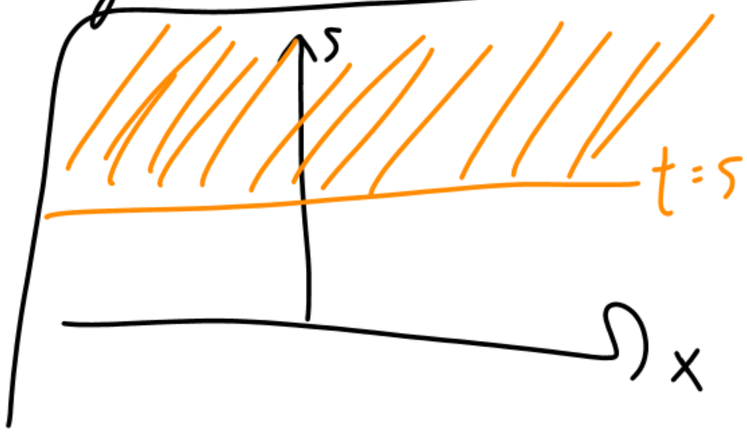
$$\begin{cases} u_t - \Delta u = f \\ u(x,0) = g(x) \end{cases}$$

$f(x,t)$ represents heat sources/sinks.

Solution formula:

First, consider $g = 0$ case.

$$\begin{cases} u_t - \Delta u = f \\ u(x,0) = 0 \end{cases}$$



Duhamel's Principle:

Idea: write

$$u(x,t) = \int_0^t v(x,t,s)\,ds$$

where u_s is the solution influenced by source term only at time t.

In detail, let $v(x,t,s)$ solve

$$\begin{cases} v_t - \Delta v = 0 & \text{in } (s,\infty)\times\mathbb{R}^n \\ v(x,t,t) = f(x,t) \end{cases}$$

Then, $u(x,t) = \int_0^t v(x,t,s)\,ds$

$$\begin{cases} v_t - \Delta v = 0, & t > s \\ v = f(x,t), & t = s \end{cases}$$

Then $u(x,t) = \int_0^t v(x,t,s)\,ds$

$$u_t = v(x,t,t) + \int_0^t v_t(x,t,s)\,ds$$

$$\Delta u = \int_0^t \Delta v(x,t,s)\,ds$$

$$u_t - \Delta u = v(x,t,t) + \int_0^t \left[v_t(x,t,s) - \Delta v(x,t,s)\right] ds$$

$$\boxed{u_t - \Delta u = f(x,t)}$$

$$u(x,0) = \int_0^0 v(x,t,s)\,ds = 0.$$

<u>Thm</u> Assume $f \in C_t^1 C_x^2([0,\infty) \times \mathbb{R}^n)$ with $D_x^2 f$, f_t bounded. Then the Duhamel solution solves

$$\begin{cases} u_t - \Delta u = f \\ u(x,0) = 0 \end{cases}$$

More useful form of Duhamel solution:

$$\begin{cases} v_t - \Delta v = 0 & \text{in } t > s \\ v(x,s,s) = f(x,s) \end{cases}$$

$$\Longrightarrow \quad v(x,t,s) = \int_{\mathbb{R}^n} \Phi(t-s, x-y) f(y,s)\, dy$$

$$\Rightarrow \quad u(x,t) = \int_0^t v(x,t,s)\, ds = \int_0^t \int_{\mathbb{R}^n} \Phi(t-s, x-y) f(y,s)\, dy\, ds$$

Convolution in (t,x) variables.